

Difficulty Key: ★ = Basic/Straightforward ★★ = Moderate ★★★ = Challenging

1 Linear Algebra Review

Problem 1. (★ VMLS 1.10 Total score.) The record for each student in a class is given as a 10-vector r , where $r_1, \dots, r_8$ are the grades for the 8 homework assignments, each on a 0 – 10 scale, r_9 is the midterm exam grade on a 0 – 120 scale, and r_{10} is the final exam score on a 0 – 160 scale. The student's total course score s , on a 0 – 100 scale, is based 25% on the homework, 35% on the midterm exam, and 40% on the final exam. Express s in the form $s = w^T r$. You can give the coefficients of w to 4 digits after the decimal point.

Problem 2. (★★ Linear independence of stacked vectors.) Consider the stacked vectors

$$c_1 = \begin{bmatrix} a_1 \\ b_1 \end{bmatrix}, \quad \dots, \quad c_k = \begin{bmatrix} a_k \\ b_k \end{bmatrix},$$

where $a_1, \dots, a_k$ are n -vectors and $b_1, \dots, b_k$ are m -vectors.

- Suppose $a_1, \dots, a_k$ are linearly independent. (We make no assumptions about the vectors $b_1, \dots, b_k$.) Can we conclude that the stacked vectors $c_1, \dots, c_k$ are linearly independent?
- Now suppose that $a_1, \dots, a_k$ are linearly dependent. (Again, with no assumptions about $b_1, \dots, b_k$.) Can we conclude that the stacked vectors $c_1, \dots, c_k$ are linearly dependent?

Problem 3. (★★ VMLS 3.5 Other common norms.) A vector norm is any real-valued function f satisfying the following four properties:

- Nonnegative homogeneity:** $\|\alpha x\| = |\alpha| \|x\|$ for all scalars α and vectors x .
- Triangle inequality:** $\|x + y\| \leq \|x\| + \|y\|$ for all vectors x and y .
- Nonnegativity:** $\|x\| \geq 0$ for all vectors x .
- Definiteness:** $\|x\| = 0$ if and only if $x = 0$.

The 1-norm and ∞ -norm for n -vectors are defined as

$$\|x\|_1 = |x_1| + \dots + |x_n|, \quad \|x\|_\infty = \max\{|x_1|, \dots, |x_n|\}.$$

Verify that the 1-norm and the ∞ -norm satisfy the four norm properties listed above.

2 K-Means

Problem 4. (★★★ Minimizing mean square distance to a set of vectors.) Let $x_1, \dots, x_L$ be a collection of n -vectors. Show that the vector z which minimizes the sum-square distance

$$J(z) = \|x_1 - z\|^2 + \dots + \|x_L - z\|^2$$

is the centroid $\bar{x} = (1/L)(x_1 + \dots + x_L)$.

(a) Explain why, for any z , we have

$$J(z) = \sum_{i=1}^L \|x_i - \bar{x} - (z - \bar{x})\|^2 = \sum_{i=1}^L (\|x_i - \bar{x}\|^2 - 2(x_i - \bar{x})^T(z - \bar{x})) + L\|z - \bar{x}\|^2.$$

(b) Explain why $\sum_{i=1}^L (x_i - \bar{x})^T(z - \bar{x}) = 0$. *Hint.* Write the left-hand side as

$$\left(\sum_{i=1}^L (x_i - \bar{x}) \right)^T (z - \bar{x}),$$

and argue that the left-hand vector is 0.

(c) Combine the results of (a) and (b) to get $J(z) = \sum_{i=1}^L \|x_i - \bar{x}\|^2 + L\|z - \bar{x}\|^2$. Explain why for any $z \neq \bar{x}$, we have $J(z) > J(\bar{x})$. This shows that the choice $z = \bar{x}$ minimizes $J(z)$.

Problem 5. (★★★ Linear separation in 2-way partitioning.) Suppose we run k-means with $k = 2$ on the n -vectors $x_1, \dots, x_N$, producing clusters with index sets G_1 and G_2 . Show that there is a nonzero vector w and a scalar v that satisfy

$$w^T x_i + v \geq 0 \text{ for } i \in G_1, \quad w^T x_i + v \leq 0 \text{ for } i \in G_2.$$

In other words, the affine function $f(x) = w^T x + v$ is greater than or equal to zero on the first group, and less than or equal to zero on the second group. This is called *linear separation* of the two groups (although *affine separation* would be more accurate).

Hint. Consider the function $\|x - z_1\|^2 - \|x - z_2\|^2$, where z_1 and z_2 are the group representatives.

3 k-Nearest Neighbors

Problem 6. (★★★ Weighted Norm Decision Boundaries.) In weighted 1-NN classification, the decision boundary between two training points $a, b \in \mathbb{R}^d$ is the set of points x satisfying

$$d_W(x, a) = d_W(x, b), \quad \text{where} \quad d_W(x, y) = \sqrt{(x - y)^\top W (x - y)}, \quad W = \text{diag}(\alpha_1, \dots, \alpha_d), \quad \alpha_i > 0.$$

- (a) Show that the decision boundary has the form of a hyperplane

$$c^\top x = d,$$

for some vector c and scalar d depending on a, b, W .

- (b) Specialize your result to $d = 2$ and write the equation of the decision boundary in terms of $\mathbb{R}^2$ coordinates x_1, x_2 .
- (c) Again with $d = 2$, show that if $\alpha_1 \neq \alpha_2$, this boundary is not perpendicular to the segment ab in the original coordinates.
- (d) Again with $d = 2$, let $z = Sx$ with $S = \text{diag}(\sqrt{\alpha_1}, \sqrt{\alpha_2})$. Show that in the z -coordinates the decision boundary is the usual perpendicular bisector of the transformed points Sa and Sb . Explain how the parameters (α_1, α_2) rescale the coordinate axes: if $\alpha_1 > \alpha_2$, distances along the x_1 direction are stretched more, so the first feature has greater influence in the weighted 1-NN classification, and similarly for $\alpha_2 > \alpha_1$.

4 Least Squares

Problem 7. (★★ VMLS 12.1 Approximating a vector as a multiple of another one.) Find a scalar x that minimizes $\|ax - b\|^2$, where a and b are nonzero m -vectors. Show that $\|a\hat{x} - b\|^2 = \|b\|^2(\sin \theta)^2$, where $\theta = \angle(a, b)$.

Problem 8. (★★ VMLS 12.7 Network Tomography.) A network has n links with unknown delays $d \in \mathbb{R}^n$. We measure travel times $t \in \mathbb{R}^N$ along N paths ($N > n$), where

$$P_{ij} = \begin{cases} 1 & \text{link } j \text{ is on path } i \\ 0 & \text{otherwise.} \end{cases}$$

Estimate the link delays $\hat{d}$ by minimizing the RMS deviation between measured times t and predicted times. Give a matrix expression for $\hat{d}$ and state any required assumptions.